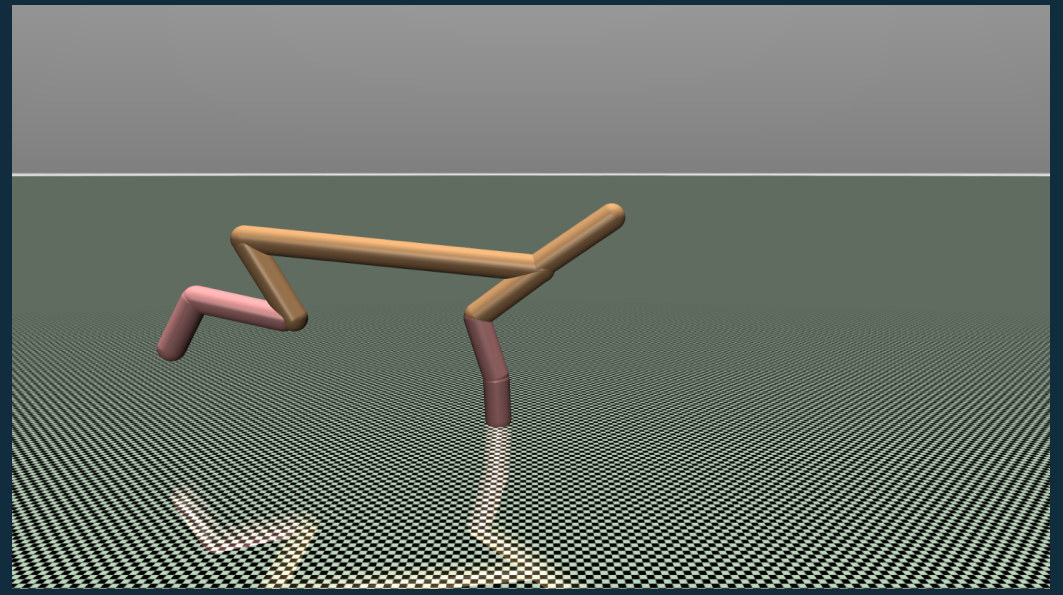
Implicit Q-Learning

HalfCheetah-v5

Offline reinforcement learning



Open notebook in Colab

GitHub repository

Aleksandr Medvedev Nikita Shankin Leo Vesin
Ruslan Nasibullin Telman Nuruzov

How IQL learns from recorded data

Offline, model-free actor-critic

Recorded data

Fixed transitions

(state, action, reward, next state)

IQL learner

Values and actor

Policy

6 bounded torques

Simulator evaluation

Complete policy episodes

Better recorded actions receive more weight

The three IQL updates

01 **Value V**

Upper expectile

→ **Favor stronger actions**

expectile = 0.7

02 **Twin critics**

Q1 and Q2

→ **reward + 0.99 × V(next state)**

Bootstrap unless truly terminated

03 **Actor**

Weighted imitation

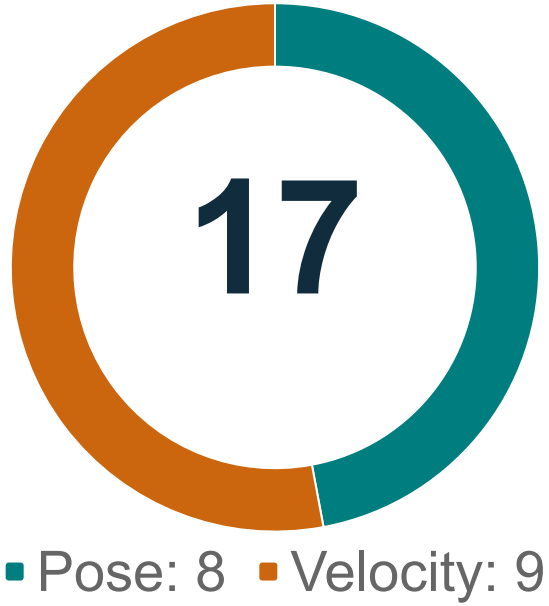
→ **min(exp(3 × advantage), 100)**

Advantage = min(target Q1, Q2) - V

Bounded policy: tanh-squashed Gaussian

HalfCheetah observations and controls

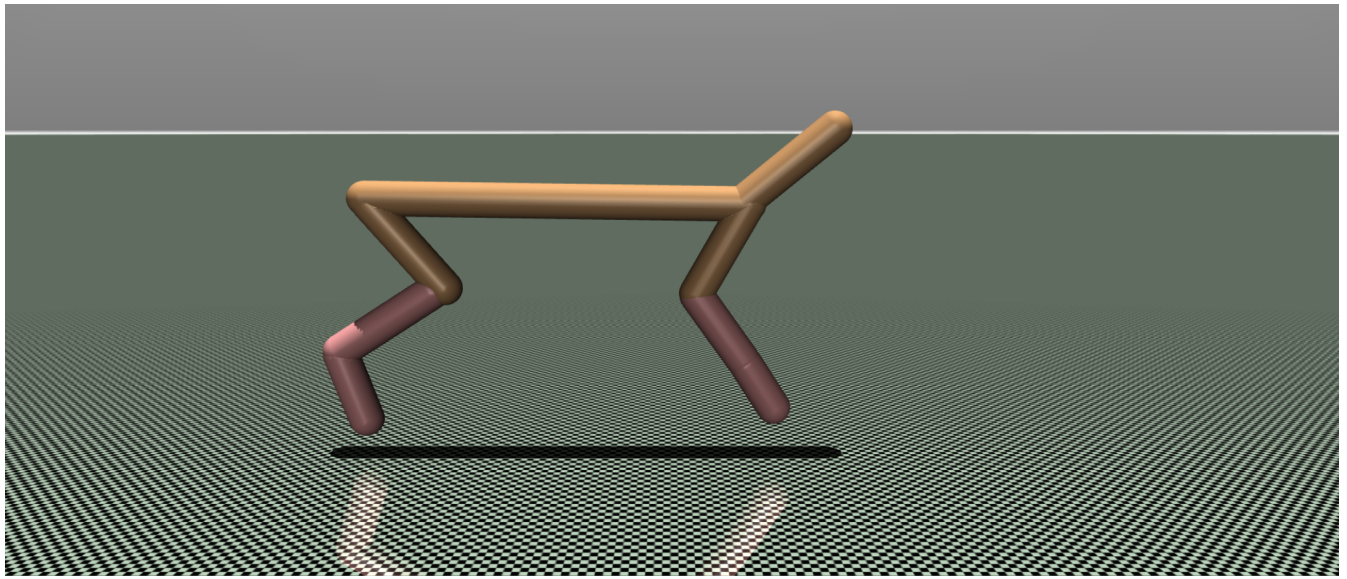
17 observations



Absolute x excluded

6 torques

[-1, 1] per joint



2 legs × (thigh + shin + foot)

Reward and episode length

Reward = forward velocity - 0.1 × sum(action²)

Example: six action components of 0.5

5.00	–	0.15	=	4.85
forward velocity		control cost		net reward

Goal: sustained, efficient forward motion

A fall does not end the episode

1,000 steps ≈ 50 seconds

Training and evaluation

1M
transitions

3
training seeds

500k
updates per seed

Minari medium: 1,000 episodes. Batch size: 256.

Train
Seeds 0, 1, 2

→

Select
10 eval episodes

→

Final test
New episode seeds

Raw return, random baseline, full videos. MuJoCo 3.2.3.

Why HalfCheetah fits IQL

Continuous torques → **Continuous actor**

Recorded locomotion → **Offline learning**

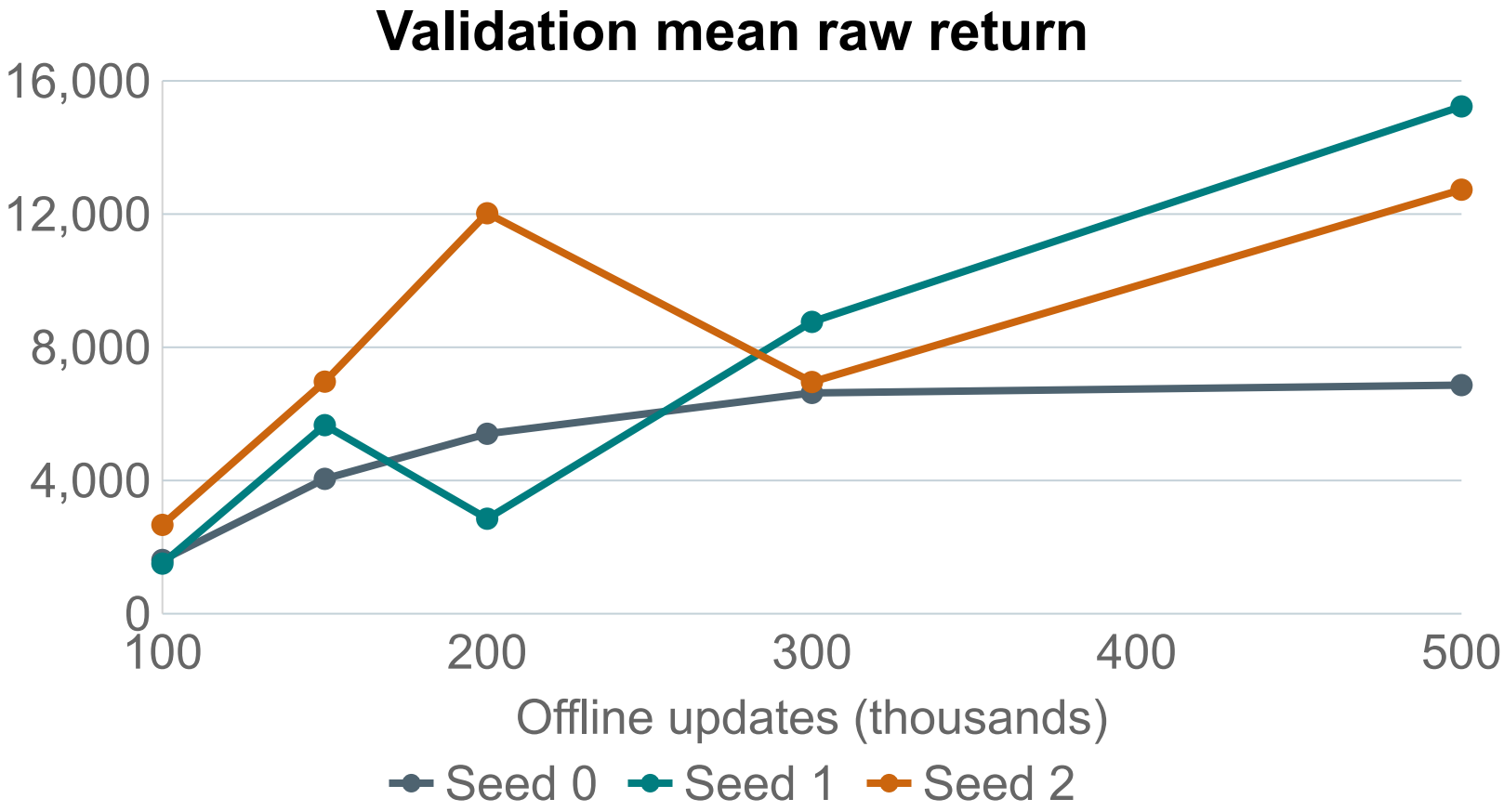
Returns and visible gait → **Measurable evaluation**

Risks: data coverage, seed failures, physics differences

Multiple seeds, complete episodes, pinned dependencies

Aleksandr: integration Nikita: data Leo: IQL
Ruslan: evaluation Telman: notebook and presentation

Learning curves and final test



15,260

final-test mean, 500k

vs 2,026 at 100k

10/10 → 0/10

episodes with long stationary tails

Means only, 10 episodes per point. Final test: seed 1, 10 new episodes.

Other training seeds still fail